CHA-3.F.1 The tangent line is the graph of a locally linear approximation of the function near the point of tangency.

CHA-3.F.2 For a tangent line approximation, the function's behavior near the point of tangency may determine whether a tangent line value is an underestimate or an overestimate of the corresponding function value.

4-6 Local Linearity and Linearization

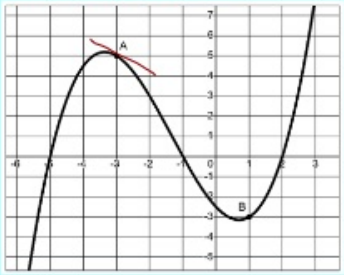
Get Ready...

Answer the following questions about the graph of $f(x)$ below.

A.) Draw the tangent line to the graph of $f(x)$ at A.

B.) Is the graph of $f(x)$ concave up or down at A?

down



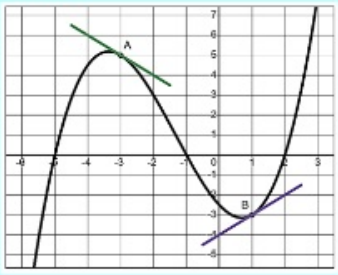
C.) How is the tangent line related to the graph of $f(x)$ at point A?

the tangent line is above the graph

D.) Draw the tangent line to the graph of $f(x)$ at B.

E.) Is the graph of $f(x)$ concave up or down at B?

Up



F.) How is the tangent line related to the graph of $f(x)$ at point B?

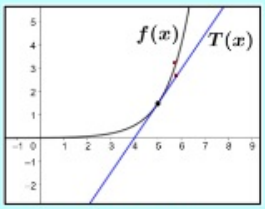
tangent line is below the graph

Recall how we write equations of tangent lines...

Writing Equations of Tangent Lines	Let $f(x)$ be a function. To write an equation of the line tangent to $f(x)$ at $x = c$, we must know two things: 1. **Point:** $(c, f(c))$ ...plug $x = c$ into the function $f(x)$ to find the point on the tangent line. 2. **Slope:** $m = f'(c)$ ...plug $x = c$ into $f'(x)$ —the derivative of $f(x)$ —to find the slope of the tangent line. Use the **Point-slope form of a line** to write the equation $y - y_1 = m(x - x_1)$ or $y - f(c) = f'(c)(x - c)$

Why tangent lines?

So, why do we even want to write equations of tangent lines? Tangent lines can help approximate values of a function when we cannot easily evaluate the function.



Notice the function $f(x)$ and tangent line $T(x)$ to the right. When $x = 5$, the tangent line $T(x)$ has the same y value as $f(x)$. At values near $x = 5$, the tangent line T(x) is *near* the graph of $f(x)$. This allows us to use $T(x)$ to approximate $f(x)$ near $x = 5$.

For example, $T(5.1) \approx f(5.1)$. This is known as the Linear approximation of f (or Linearization or Tangent line approximation) at $x = 5.1$.

Since $f(x)$ is concave up at $x = 5$, the tangent line, $T(x)$ is below the graph of $f(x)$. This means our approximation of $f(5.1)$ is an **underestimate**.

If $f(x)$ were concave down at $x = 5$, the tangent line, $T(x)$ would be above the graph of $f(x)$. In this situation, our approximation of $f(5.1)$ would be an **overestimate**.

Example 1: The functions f and g are differentiable with selected values of , g, f', and g' given above.

Let $h(x) = f(g(x))$.

x	1	4	5
$f(x)$	-4	0	-9
$g(x)$	6	5	8
$f'(x)$	2	-3	3
$g'(x)$	-5	-2	7

$h(x)=f(g(x))$

a) Write an equation of the line tangent to $h(x)$ at $x = 4$.

$h(4)=f(g(4))=f(5)=-9$

$h'(x)=f'(g(x))\cdot g'(x)$

$\text{Slope}=h'(4)=f'(g(4))\cdot g'(4)$

$=f'(5)\cdot g'(4)$

$=3(-2)=-6$

$EQ_{TAN}: y+9=-6(x-4)$

b) Use the tangent line from part (a) to approximate $h(\overset{x}{4.1})$.

$\underbrace{h(4.1)}_{y}+9\approx-6(4.1-4)$

$h(4.1)\approx-9-6(4.1-4)$

$=-9-6(.1)$

$=-9-.6=-9.6$

$h(4.1)\approx -9.6$

Example 2: Let $y = f(x)$ be a function with $\frac{dy}{dx} = \frac{3x}{(y+4)^2}$ and containing the point $f(1) = -5$. It is known that $f(x)$ is concave up at $x = 1$.

(a) Write an equation for the line tangent to the graph of $y = f(x)$ at $x = 1$. Use your equation to approximate $f(0.8)$.

$f(1) = -5$

$\text{Slope} = \left.\frac{dy}{dx}\right|_{(1,-5)} = \frac{3}{(-1)^2} = \frac{3}{1} = 3$

$EQ_{TAN}: y + 5 = 3(x-1)$

$f(0.8) \approx -5.6$

$f(0.8) \approx -5 + 3(0.8-1) = -5 + 3(-0.2)$

$= -5 - 0.6 = -5.6$

(b) Is your answer in part (a) an overestimate or underestimate? Explain.

Underestimate because $f(x)$ is concave up at $x=1$.

Example 3: On Thanksgiving, a delicious pumpkin pie removed from an oven has temperature 175°F. The temperature of the pie at time t minutes can be modeled by the function P where $\frac{dP}{dt} = -\frac{1}{50}(P - 25)$. Use the tangent line to P at $t = 0$ to approximate the temperature of the pie 4 minutes after being removed from the oven.

$P(0) = 175$

$\text{Slope} = \left.\frac{dP}{dt}\right|_{(0,175)} = -\frac{1}{50}(175-25) = -\frac{1}{50}(150) = -3$

$EQ_{TAN}: y - 175 = -3(t - 0)$

$P(4) \approx 175 - 3(4) = 175 - 12 = 163°F$